Chapter 8

Dispensing Process, Medication Reconciliation, Patient Counseling, and Medication Adherence

Ema Paulino[1], Dixon Thomas[2], Shaun Wen Huey Lee[3] and Jason C. Cooper[4]

[1]Pharmacy, Nuno Álvares, Almada, Portugal; [2]Gulf Medical University, Ajman, United Arab Emirates; [3]Monash University Malaysia, Bandar Sunway, Malaysia; [4]Medical University of South Carolina, Charleston, SC, United States

Learning Objectives

Objective 8.1 List steps in the dispensing process.
Objective 8.2 Detail methods in medication reconciliation to prevent errors.
Objective 8.3 Explain approaches to patient counseling.
Objective 8.4 Outline strategies to improve medication adherence.

OBJECTIVE 8.1. LIST STEPS IN THE DISPENSING PROCESS

Good dispensing practices ensure that an effective form of the correct drug is delivered to the right patient, in the correct dosage and quantity, with clear instructions, and in a package that maintains the potency of the drug. Dispensing also includes all of the activities that occur between the time the prescription is presented to the pharmacy and the time the drug or other prescribed items are issued to the patient.[1]

Dispensing is a fundamental service that pharmacists have been traditionally providing. Although non-prescription drugs and products can be dispensed following a short interview or physical assessment of the patient, prescription-only drugs are mostly dispensed as per prescriber's instructions. Besides community pharmacies, certain prescription-only drugs may be dispensed by hospital pharmacy outpatient services in a significant number of countries.[2] In fact, in some countries, especially in Asia, prescription-only drugs are mainly dispensed by hospital pharmacies, as is the case in China, Indonesia, or Taiwan. In some jurisdictions, pharmacists are allowed to independently prescribe and dispense from a limited list of drugs. Most jurisdictions reserve the right and responsibility to dispense drugs exclusively to pharmacists, whereas some grant dispensing privileges to doctors and/or nurses.

Supply chain systems aim to promote the continuous availability of the right drug, at the right time, to the right patient. To that end, supply chain integrity and efficiency is ensured by several stakeholders throughout the distribution pathway, including manufacturers, wholesalers, and pharmacies.

Substandard and falsified drugs threaten patient safety by, at best, causing no improvement or, worse, causing added burden of disease and even death; endangering public health by increasing the risk of antimicrobial resistance; and eroding patients' trust in health professionals and health systems. As part of the healthcare system, pharmacists and other healthcare professionals have a shared responsibility to avoid penetration of such drugs within the legitimate pharmaceutical supply chain. The International Pharmaceutical Federation (FIP), in cooperation with the Indian Pharmaceutical Association, has developed a handbook for healthcare professionals entitled *All You Need to Know about Spurious Medicines*, which contains tips on preventing and detecting substandard and falsified drugs; additionally, it contains advice on how to

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minimize the threat and change behaviors.[3] This handbook was launched as part of a World Health Professions Alliance campaign and aimed to raise awareness and develop competencies of healthcare professionals in the prevention of substandard and falsified drugs in the supply chain. It also serves to enhance the detection of and the ability to communicate such issues to patients and other stakeholders.

For instance, to address the concerns about increasing falsified drugs, the European Union amended the Directive 2001/83/EC by implementing measures to prevent the entry into the legal supply chain of falsified drugs. Such requirements involved a unique identifier and an anti-tampering device on the package of drugs for human use. This system consists of an end-to-end verification of those drugs, which means that pharmacists have to scan the box and verify the product against a central database at the point of dispensing. If located within the system, that product is then recognized and considered legitimate.

To avoid issues around the quality of drugs that are dispensed, guaranteeing that the right storage conditions are kept is of utmost importance. Pharmacies also often hold obligations in reverse logistics, ensuring a safe medication waste-management program and, if necessary, recall and withdrawal operations in case the pharmacist identifies a medical product that may be substandard or falsified. In the latter example, pharmacists often hold the obligation to report this to the relevant authorities and should do so even when this obligation is not in place.

Dispensing of controlled drugs requires more documentation and accountability to prevent drug diversion and potential abuse. Opioid overdose-related deaths are increasing; therefore, pharmacists' interventions are required on appropriate opioid use, with special provisions related to narcotic/opioid dispensing regulations implemented worldwide. Opioid stewardship programs are also in place to ensure judicious use of opioids.[4,5]

Antimicrobial stewardship programs require prudent prescribing and dispensing of antimicrobials. Overprescribing, inappropriate use, and dispensing without a prescription are reasons for unnecessary use of antibiotics that eventually lead to increased antimicrobial resistance. Healthcare—associated infections are often caused by resistant microorganisms. Optimizing prescribing and dispensing practices decreases antimicrobial misuse, which results in better patient outcomes at a lower cost for the individual and the healthcare system.[6,7] Specific emphasis should be given to collaborative approaches as part of interprofessional teams or interdisciplinary management of antimicrobial strategies along with stakeholder engagement.[8]

Pharmacists are responsible for ensuring that appropriate medications are prescribed, dispensed, and administered to the patient or their carer.[9] The steps involved in the dispensing of drugs are described in Fig. 8.1.[1,10] This should be consistent with the relevant national drug policy of the respective country. Good Pharmacy Practice guidelines describe in further detail the best practices in dispensing.[11]

In dispensing, a systematic approach is required, preferably following a written protocol; thus, there is a need for clear documentation and risk management procedures to minimize the possibility of any error that can arise during any stage of the dispensing process.

Receive and Validate the Prescription

The first step in dispensing is to validate the prescription. This is needed to ensure that

- details are accurate and complete, including the patient's name;
- legal and regulatory requirements are satisfied.

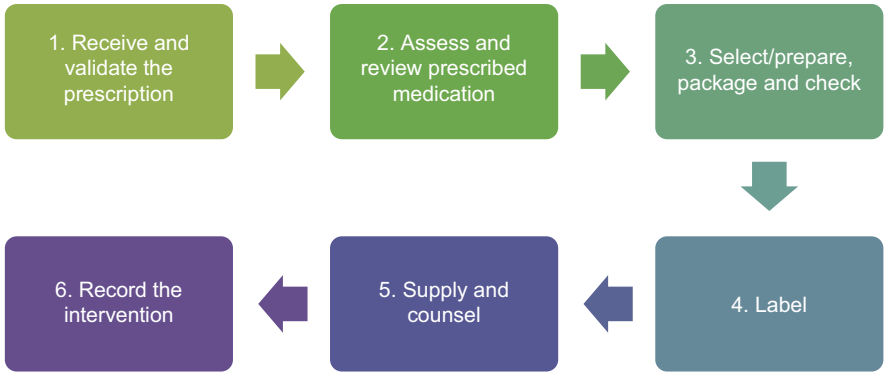


FIGURE 8.1 Steps in dispensing drugs safely.

Assess and Review the Prescribed Drugs

In this step, pharmacists assess the prescription, interpret the prescribed drug, and check whether the dose, frequency, route, and quantity are appropriate for the intended patient. The pharmacist also quickly assesses the patient and should adapt his/her intervention accordingly.

While assessing the prescription, a pharmacist should systematically collect and accurately record any relevant information (e.g., comorbidities, allergies, use of complementary medications). This is especially important in groups of people who are at risk of medication-related issues, including those patients[12,13]

- recently discharged from the hospital or with multiple transitions of care;
- taking multiple drugs (five or more);
- with complex chronic conditions;
- taking drugs with a narrow therapeutic index;
- taking drugs that have increased potential to interact with other drugs.

The pharmacist should further consider the appropriateness of the prescribed drug, considering the patient's clinical record and dispensing history based on the following principles:

- medication safety and efficacy, considering patient-specific factors such as age, allergies, other health conditions, and pregnancy/lactation;
- dosage regimen;
- dosage form;
- potential interactions;
- contraindications;
- precautions;
- patient's medication experience, needs, and adherence to the prescribed regimen, including misuse and abuse issues (which may be intentional or unintentional).

In particular, some medications have a high risk of causing death or harm. These include drugs with a narrow therapeutic index. Dosing errors with these medications can result in serious consequences. These drugs are often represented by the acronym APINCH[14]:

- **A**nti-infective;
- **P**otassium and other electrolytes;
- **I**nsulin;
- **N**arcotics and other sedatives;
- **C**hemotherapeutic agents;
- **H**eparin and other anticoagulants.

Although this list is nonexhaustive, it can be a good guide to alert pharmacists for some of the more commonly prescribed drugs, which may present an increased risk for harm.

The prescriber should be contacted whenever there is a need to discuss the patient's prescription or therapeutic regimen. This is also an opportunity to collaborate with the prescriber and contribute to clinical decision-making. Some common issues a pharmacist would contact a prescriber include the following:

- concern about the suitability of the drug;
- concern about the potential for medication-related problems;
- doubt about the legality or validity of the prescription;
- uncertainty about the prescriber's intentions.

It is important that pharmacists prioritize and list some of the key concerns during the discussion. In addition, the pharmacist can offer evidence-based strategies or solutions to the prescriber to promote optimal medication use. Always document any outcome of the discussion with the prescriber and inform the patient of the result of the discussion. In the event that the prescriber is unavailable, or unwilling to accept the pharmacist's advice, use professional judgment to determine the most appropriate action to take to satisfy the duty of care to the consumer.

Select/Prepare, Package, and Check Against the Original Prescription

In this step, pharmacists select the appropriate drug, brand, strength, form, and quantity. In countries where this applies, the pharmacist may need to repack. When applicable, the pharmacist should prepare the drug, which may include reconstituting (e.g., antibiotics) or compounding from raw materials.

In some instances, the pharmacist might have to deal with requests for out-of-stock drugs. Medication shortages have become a global issue, putting patient lives at risk and creating difficulties for healthcare professionals.[15] The causes of these shortages are multidimensional in the context of a complex global supply chain and thus require a holistic approach. In some jurisdictions, pharmacists may be able to independently substitute for a therapeutic alternative. When this is not the case, the pharmacist should contact the prescriber to ensure that the patient receives the medication needed. In addition, when faced with a shortage, pharmacists should report this to the competent authority or the appropriate organization, such as the American Society of Health-System Pharmacists in the United States. Reporting shortages is important to support public health decisions, provide guidance and information to enable all healthcare professionals to limit the impact of such shortages, and offer data to monitor the impact of public health policies that aim to curb this issue.[15]

Pharmacists should, at this stage, make a visual inspection of the drug. This visual inspection allows for the pharmacist to detect signs of potential falsified drugs such as improper packaging, labeling, and description of dosage, and to check whether the product is in good dispensing conditions (e.g., expiry date, integrity of the container/box). FIP, together with the US Pharmacopeia and the International Council of Nurses, developed a checklist for a visual inspection of drugs to help identifying suspicious products for further examination.[3]

This is also a good opportunity for the pharmacist to discuss with the patient any generic substitution. It is important for the pharmacist to clarify with the patient on their knowledge, attitudes, and preferences toward the use of a generic product versus the brand-name medication. This can be an opportunity for the pharmacist to identify any real or potential confusion related to generic/brand substitution.[16]

The final portion of this step should include a dispensing check against the original prescription. This should involve a visual check or a scan of the drug barcode, if available. Scanning the barcode also allows for documentation of the dispensing.

Label Drugs

Always prepare and attach any dispensing and cautionary advisory labels in a way that best meets the needs of the patient. Some considerations include the following:

- medical conditions;
- eyesight;
- health literacy.

Always ensure that the drug labeling is accurate, unambiguous, legible, and complete, as any misunderstandings can lead to unintentional misuse and poor health outcomes. Always follow the best practice for labeling drug containers[17]:

- Use dark print and a suitably sized font;
- Attach dispensing labels and appropriate cautionary advisory labels to the primary container (e.g., bottle or inhaler) to ensure that the directions are readily available;
- Comply with relevant legislation and guidelines;
- Attach labels to flat surfaces (i.e., not over the edge of the box);
- Use purpose-designed tags or “winged” labels if the primary container is very small;
- Ensure that the dosage and name of the active ingredient(s) are visible;
- Ensure that the batch number, expiry date, and storage requirements are visible;
- Avoid placing labels over barcodes or seals.

Supply and Provide Counseling

Counseling is a vital step in the dispensing process. This is to ensure that the patient will have sufficient information to understand their drugs and intended therapeutic effects. Always give the patient the opportunity to ask any questions and clarify any information. In counseling, discuss the following:

- what the drug is used for;
- how the drug works;
- how to take drug, use devices, and measure doses;
- when to take the drug;
- how long to continue taking the drug;

- what to do if a dose is missed;
- what to do if a dosing error or overdose occur;
- how to recognize adverse effects and what to do if they occur;
- how to store or dispose of the drug;
- lifestyle advice relevant to the condition or disease (e.g., smoking, alcohol);
- potential interactions with other drugs, complementary drugs, or foods.

Please bear in mind that different patients may have different informational needs; the intervention should be customized accordingly.

Always highlight any drugs and brands that are new to the patient using an appropriate language. Written information can be useful to supplement verbal counseling.

Record the Intervention

Pharmacists should keep a record of the prescribed and dispensed medication, as well as of any interventions undertaken by the pharmacist.

This record of drugs dispensed may be constructed automatically through scanning the drug box or barcode, which also allows for correct billing.

When patient records are available, the possibility to create and maintain the medication history of that specific patient has several benefits, including the ability to intervene and prevent polypharmacy, drug interactions, and/or medication errors.

OBJECTIVE 8.2. DETAIL METHODS IN MEDICATION RECONCILIATION TO PREVENT ERRORS

It is important for a pharmacist to reconcile a patient's current prescription with the medication history obtained. This usually involves comparing the prescribed medication with medication history, identifying and resolving any discrepancies that may arise, and documenting the changes in the medication regimen. Prescribing, dispensing, and administration of new drugs should be performed after proper reconciliation, as an accurate medication history is essential to avoid medication errors.[18] It has been found that availability of an electronic database, which includes patient medication history, enhances on-admission medication reconciliation process.[19]

Studies have shown that poor, or lack of, medication reconciliation constitutes a significant risk for medication discrepancies, errors, and adverse drug reactions that can result in adverse events, particularly when transitions of care occur.[20,21] In a systematic review and metaanalysis published in 2016, pharmacy-led medication reconciliation interventions were found to be an effective strategy to reduce medication discrepancies and had a greater impact when conducted at either admission or discharge.[22]

Medication reconciliation is not only relevant at transitions between different levels of care but also in the transfer of care within the hospital. Medication errors increase when intensive care unit (ICU) transfers are high. Pharmacists' medication reconciliation was found to decrease medication transfer errors at ICU admission and prior to ICU discharge.[23] In addition, pharmacists demonstrate their advocacy roles throughout the transition of care on behalf of the patient. Pharmacists are empowered with patient advocacy skills to prevent medication errors through medication reconciliation. Fig. 8.2 shows a pharmacist verifying a medication order.[24]

A standard operating protocol (SOP) for medication reconciliation was developed, tested, and refined for use within the context of the WHO Action on Patient Safety ("High 5s") initiative. This is an internationally coordinated, limited participation activity for testing the feasibility of implementing standardized patient safety protocols and determining the impact of the implementation on certain specified patient safety outcomes.[25] It applies to the acute hospital setting and covers implementing medication reconciliation on admission, at internal transfer, and on discharge from hospital. This SOP established guiding principles for implementation of this process, namely:

1. An up-to-date and accurate patient medication list is essential to ensure safe prescribing in any setting;
2. A formal structured process for reconciling medications should be in place across all interfaces of care;
3. Medication reconciliation upon admission is the foundation for reconciliation throughout the episode of care;



FIGURE 8.2 Pharmacist verifying a prescription. *Courtesy https://commons.wikimedia.org/wiki/File:US_Navy_030819-N-9593R-082_Pharmacist_Randal_Heller,right_verifies_the_dosage_and_medication_of_a_prescription_at_the_National_Naval_Medical_Center_in_Bethesda,_Maryland.jpg.*

4. Medication reconciliation is integrated into existing processes for medication management and patient flow;
5. The process of medication reconciliation is one of shared accountability with staff being aware of their roles and responsibilities;
6. Patients and families are involved in medication reconciliation;
7. Staff responsible for reconciling medications are trained to take a patient's list of current medications and reconcile with previous drugs/regimens.

Medication reconciliation also occurs in community pharmacies and is especially important for patients who are discharged from hospitals or other healthcare facilities and/or who are visiting more than one physician.[26–28] Further research on clinical relevance of discrepancies and potential benefits on reducing healthcare team workload should be undertaken.[29]

OBJECTIVE 8.3. EXPLAIN APPROACHES TO PATIENT COUNSELING

The community pharmacy is often the first point of contact between patients and the healthcare system. Pharmacists provide education and pharmaceutical care to help patients to understand and use drugs responsibly.[30] Counseling is part of the dispensing process and includes instructions on how to administer medications and what to do if something goes wrong.[9]

Patient education is usually a broader term and has been described as an element of patient empowerment. This can be achieved by teaching patients about their illnesses/conditions and encouraging greater involvement in decisions related to ongoing care and treatment.[31]

The primary purpose of offering drug information to patients is to improve public health literacy and to empower and assist citizens in achieving safe, effective, and appropriate use of drugs. This includes providing information that allows users of drugs to make informed decisions about their healthcare.[32] Users of drugs value accurate, comprehensible, appropriate, objective, independent, up-to-date, and relevant information, which can adequately inform and assist them in safe, effective, and appropriate self-management.

Pharmacists influence drug therapy, public health, and disease prevention through counseling. Specialized disease counseling improves patient outcomes. Although outcomes may vary in different health systems and individual patients, counseling has been found to be effective.[33,34]



Patient Counseling on Over-The-Counter medication. *Courtesy https://commons.wikimedia.org/wiki/File:Pharmacist_assisting_a_customer_at_Terry_White_Chemists.jpg*.

Pharmacists also counsel consumers on health promotion and disease prevention, even when a drug is not dispensed. Depending on practice settings, job description, and patient needs, counseling provided by pharmacists may vary. In fact, there is much evidence demonstrating that pharmacist input into self-care is highly effective. The value of this pharmacist input derives from[35]

- Professional competency: The ability to safely assess minor illnesses effectively and distinguish them from major disease;
- Economic factors: The ability to support self-care efficiently by reducing health costs, both in terms of drug and salary costs and through indirect costs (e.g., enabling people to remain at work, minimizing time off work);
- Integration factors: The ability to ensure continuity of care; and
- Communication and access factors: The ability to interact effectively with the public.

Whenever counseling is used as an intervention (e.g., to improve awareness and medication adherence), pharmacists ideally use the five-step approach in Pharmacists' Patient Care Process:[36]

1. Information is collected on nonadherence of patient to the medication;
2. The pharmacist assesses the reasons for the nonadherence;
3. Together, with the patient and other pertinent healthcare professionals, the pharmacist prepares a care plan, which in the case of nonadherence, may be to clarify doubts related to medication use;
4. The pharmacist implements the care plan in collaboration with other healthcare professionals and the patient or caregiver;
5. The pharmacist monitors changes in adherence patterns and evaluates the effectiveness of the intervention.

Counseling in acute and chronic disease conditions is different. Counseling in certain diseases needs special training (e.g., diabetes, HIV/AIDS, psychiatric). For instance, chronic conditions are prone for deviations in medication adherence patterns due to many factors. Conditions with social stigma require a private and comfortable counseling environment.[37] Care should be taken while counseling patients with dementia or sensory impairments. Appropriate counseling aids may be required to effectively communicate the message.[38]

Counseling for specific populations also needs a varied approach. The approach to pediatric, geriatric, or pregnant populations should be different according to common issues and concerns within that population. For instance, when drugs are intended for children, their caregiver should be involved, as nonadherence to drugs is an important issue for pediatric patients. To ensure compliance, the pharmacist must optimize a child's therapy regimen by partnering with the family to identify barriers to compliance, provide education on the importance of compliance, provide strategies to help children take their medications, and offer ongoing support and assistance. By involving the child in this process, the pharmacist must communicate in an age-appropriate manner, utilizing appropriate aids when necessary.[39]

Geriatric patients can pose significant challenges to counseling by the pharmacist. Complex medication regimens, physical limitations, cognitive impairment, economic issues, adherence, and attitudes toward sickness and medications should be addressed. This may necessitate the counseling process to occur with a caregiver or family member.[40]

Pregnant and lactating mothers are concerned about the safety of their fetus/infant. Addressing the probability of harm for each drug is therefore paramount. Specialized references (e.g., *Briggs Pregnancy and Lactation*) should be consulted to provide accurate information.

Extremely careful strategies are required while counseling a noncooperating, angry, or frustrated patient. The pharmacist should apply active listening techniques and address patients' concerns and causes for frustration in a professional, assertive, but empathic manner. If a patient refuses counseling, the pharmacist should ascertain whether this refusal is based on the patient's conviction that they already know how to handle their medication. In this case, asking how the patient has been taking their drugs and how they are feeling might provide the opportunity to identify any knowledge gaps and address those in a constructive manner.

It is important to acknowledge that patients have the right to accept or deny advice and counseling. Counseling should not be imposed but offered. It should be provided at the right extent through an approach that is acceptable to consumers. Understanding cultural sensitivity is essential; a comment that negatively addresses culture or beliefs could result in nonadherence. Studying and practicing in a community with diversity is associated with better outcomes. The Institute of Medicine has stated that greater diversity within healthcare professionals leads to improved patient outcomes.[41]

Steps in the patient education and counseling process may vary according to needs of the individual, environment, and practice setting. The following is a generalized list of things to remember when counseling patients[40,42]:

- Establish a relationship that will maximize effective communication by demonstrating genuine interest, acceptance, and rapport.
- Address people using their preferred name.
- Introduce yourself as a pharmacist, explain the purpose and expected length of the session, and obtain the patient's agreement to participate.
- Determine patient-specific barriers to communication and implement a strategy to overcome barriers.
- Assess the patient's knowledge about health problems and medications, physical and mental capability to use the medications appropriately, and attitude toward the health problems and medications used.
- Provide information orally and use visual aids or demonstrations to fill the patient's gap in knowledge and understanding. Show the patient the colors, sizes, shapes, and markings on oral solids. For oral liquids and injectable drugs, show patients the dosage marks on measuring devices. Demonstrate the assembly and use of administration devices such as nasal and oral inhalers. As a supplement to face-to-face oral communication, provide written handouts to help the patient recall the information.
- Use active listening skills, good eye contact, and gestures when appropriate.
- Observe nonverbal cues such as body language, behavior or facial expression, for reactions.
- Give support, encouragement, and feedback.

When dealing with complaints, the method used by Starbucks employees is worth mentioning, LATTE. LATTE stands for "We **Listen** to the customer, **Acknowledge** their complaint, **Take** action by solving the problem, **Thank** them, and then **Explain** why the problem occurred."[43]

OBJECTIVE 8.4. OUTLINE STRATEGIES TO IMPROVE MEDICATION ADHERENCE

Nonadherence occurs when patients do not take their drugs appropriately or at all. Studies have shown that approximately 50% of patients do not take medications as prescribed.[44] Nonadherence is driven by a combination of factors versus just one. These include lack of affordability; unintended patient-related factors such as forgetfulness or an unsupportive patient and healthcare professional relationship; or inappropriate patient−product suitability such as packaging/device, complexity of medication regimen, and adverse drug reactions. All of these affect patient responsiveness and drug use.[44,45]

Poor adherence can impair the efficacy and safety of drugs, reduce the full benefits of treatment, and lead to unnecessary adverse events and hospitalization. This usually results in costly complications, which are often more expensive than drugs, and poorer health outcomes.[46]

Some authors argue that the terms compliance and adherence have differences, whereas some use it interchangeably.[44] Compliance is the extent to which the patient's behavior matches the prescriber's recommendations. Adherence is the extent to which the patient's behavior matches the agreed-on recommendations from the prescriber. Many prefer to use the term adherence over compliance, because this later implies patient passivity. Concordance is a similar but broader term that

stresses patient support in drug taking. Generally, the goal is achieved if the patient takes the medication as recommended to achieve maximum therapeutic benefit.[47] Adherence is the more popular term used.

Improving medication adherence is a challenge. Pharmacists are not in control of many factors that decrease medication adherence. However, being a drug expert and the last professional in the patient care process, pharmacists have leading responsibilities to implement strategies and services which enhance medication adherence.

Initial medication adherence (IMA) and persistence are different in certain aspects. IMA is defined as the patient obtaining, for the first time, a new prescription medication. Difficulties in access to a prescribed drug due to nonavailability in the pharmacy or the drug being too costly are issues leading to nonadherence. More complex patient factors may exist, however, as many patients may not use their medications from the very beginning for a variety of reasons.[48] Medication persistence has been defined as "the duration of time from initiation to discontinuation of therapy."[49] Persistence is particularly relevant in the context of chronic diseases and may require tailored interventions, as adherence changes over time.

Counseling and providing instructions with assertiveness is important to win the trust of patients and to motivate them to take their drugs as instructed. Pharmacists should not contradict prescriber's instructions, as this may decrease trust between the healthcare provider and patient. Any intervention to modify the prescription should be made in collaboration with the prescriber, to avoid losing the trust of the patient.

Many methods are in place to improve medication adherence, and these relate to the reasons for nonadherence.[44] To identify the best strategy or service for a given situation, it is important to determine if the nonadherence is intentional or unintentional.[50] Unintentional nonadherence (e.g., poor memory, work restrictions, mental illness) is usually more easily resolved and could involve simple adjustments to the medication regimen or even providing certain services such as medication administration aids (e.g., medication calendars, pill organizers).

Intentional nonadherence requires a more structured approach, as reasons usually involve patients' beliefs, concerns, or issues related to side effects of drugs, fear of dependency, or mistrust in the benefit.

It is important to create a "shame-free" environment to address poor health literacy and ask questions in the affirmative.[44,50] Patients should not feel ashamed or feel that they have done something wrong. Otherwise, they might stop revealing nonadherence issues with the pharmacist, with fear that there is an unpleasant response.

A health coaching or motivational interviewing approach is usually most suitable for a patient counseling session. This takes into account not only the pharmacist being an expert in medication use but the patient also being an expert on their health and personal circumstances which can influence medication decisions.

The pharmacist could adopt the following approach for a counseling session:[44,51–53]

1. Explore the patient's perspective;
2. Address any beliefs and concerns the patient may have which resulted in the nonadherence;
3. Explain key information when prescribing/dispensing a drug, including key information about the drug;
4. Share and discuss the advantages and disadvantages of different drugs options, including common side effects;
5. Reach a shared understanding about the issues that affect adherence;
6. Use, when applicable, medication adherence-improving aids, such as medication calendars or schedules that specify the time to take medications, drug cards, medication charts; drug-related information sheets; or specific packaging strategies such as pill boxes, "unit-of-use" packaging, and special containers indicating the time of dose;
7. Provide behavioral support to incorporate the medication regimen into the patient's daily regimen;
8. Negotiate, agree, and record the plan;
9. Schedule appropriate follow-up to assess the medication adherence and identify further difficulties and barriers to be addressed.

There are several methods to measure adherence, which can be broken down into direct and indirect methods of measurement.[44,52] Direct methods include direct observed therapy, measurement of the level of a drug or its metabolite in blood or urine, and detection or measurement of a biological marker added to the drug formulation. Direct approaches are most accurate but are more expensive. Indirect methods include patient questionnaires, patient and caregivers' self-reports, pill counts, rates of prescription refills, assessment of patient's clinical response, electronic medication monitors, measurement of physiologic markers, and patient diaries.

A recent systematic review addressing the effectiveness of interventions by community pharmacists to improve patient adherence to chronic medications concluded that, although counseling, monitoring, and education during weekly or monthly appointments showed some effect, it was impossible to identify an overall successful adherence-improving strategy performed by pharmacists.[54] The authors concluded that more well-designed and well-conducted studies need to be performed.